

RUSHENDRA SIDIBOMMA

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🌐 [rushendra-sidibomma](https://rushendra-sidibomma.github.io)

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EDUCATION

University of Minnesota Twin-Cities

MS Computer Science (GPA: 3.89/4)

Sep 2025 – May 2027 (Expected)

Minneapolis, MN

Indian Institute of Information Technology, Sri City

Bachelor of Technology in Computer Science with Honours (GPA: 9.57/10)

Dec 2020 – July 2024

Sri City, India

TECHNICAL SKILLS

Programming and Scripting Languages: Python, C/C++, JavaScript, SQL, Bash

Backend and APIs: Django, FastAPI, RESTful API design, Microservices, Authentication, Async Processing

Databases and Infrastructure: PostgreSQL, Distributed Systems, Schema Design, Linux, AWS, Azure, Docker, CI/CD

ML Systems: Model Serving and Deployment, Inference Optimization, PyTorch, LangChain, Transformers, Ollama

EXPERIENCE

AutomationEdge Technologies

Backend Engineer (ML Systems)

July 2024 – July 2025

Pune, India

- Built and deployed a **document intelligence backend service**, owning API design and the core system architecture.
- Implemented **15+ REST API** endpoints using Django to support secure document ingestion, orchestration, and structured results delivery for **enterprise customers**.
- Developed **asynchronous background processing** pipelines to handle large medical documents (300+ pages), achieving end-to-end processing times under 90 seconds.
- Deployed** the service in production on **AWS EC2** with centralized logging and monitoring in a high-availability, multi-VM setup.
- Scaled the backend service to handle **1000+** daily requests, reducing patient intake errors by **30%** and operational costs by **60%** compared to prior manual workflows.
- Extended the backend service for multi-tenant deployments; live in **production across 10+ home health agencies** in the U.S., generating over **\$200K** in annual recurring revenue.

Eyemote Vision

Backend Engineer Intern (ML Systems)

Aug 2023 – Jan 2024

Remote

- Built a backend inference service to detect and grade the severity of diabetic retinopathy from fundus images by exposing model predictions for clinical workflows.
- Deployed the models as cloud-based inference services optimized for **low-latency prediction**, enabling near **real-time evaluation** on incoming patient data.
- Integrated** the inference pipeline with partner eye clinics' internal systems, supporting end-to-end data flow from image ingestion to model predictions in **production settings**, and shipped the solution as a **minimum viable product (MVP)**.

PROJECTS

Multi-Tenant API Gateway with Rate Limiting | Django, Redis, PostgreSQL

- Built a backend **API gateway** enforcing per-tenant rate limits using **token-bucket throttling** to ensure fair usage across clients.
- Designed tenant-aware request tracking with persistent counters and burst handling under **concurrent workloads**.
- Implemented graceful degradation and standardized error responses for throttled and failed requests.

Knowledge Management Service | Django, REST APIs, RAG

- Built a **multi-tenant** backend service exposing REST APIs for document ingestion, query handling, and response generation for knowledge retrieval.
- Designed and implemented an **asynchronous** document ingestion and indexing pipeline, improving end-to-end query response latency by **20%** under concurrent workloads.
- Deployed and operated the service in production, improving query resolution rates by **15%** under real user traffic.

Intelligent Email Ticketing and Service Automation System | Django, LLMs, Inference Optimization

- Built an on-prem ticket classification and routing service using a lightweight LLM distilled from enterprise models, enabling **low-latency CPU inference** for 20+ IT service request categories.
- Integrated the inference service into an **automated ticket handling** pipeline, eliminating manual triaging and reducing issue resolution time by over **60%** while handling **500+ daily requests**.

PUBLICATIONS

- M. Maleki, **R. Sidibomma**, A. Adibi, R. Samavi. **Provable Robustness with SOCP: A scalable Approach to Certified Verification.** (Submitted)
- M. Maleki, **R. Sidibomma**, A. Adibi, R. Samavi. **Cascading Robustness Verification: Toward Efficient Model-Agnostic Certification.** ([Link](#)) (Accepted at SaTML 2026)
- **Sidibomma, R.**, Patwa, P., Patwa, P., Chadha, A., Jain, V. and Das, A., 2025, January. LLMsAgainstHate@ NLU of Devanagari Script Languages 2025: **Hate Speech Detection and Target Identification in Devanagari Languages via Parameter Efficient Fine-tuning of LLMs.** In Proceedings of the First Workshop on Challenges in Processing South Asian Languages (CHIPSAL 2025) (pp. 301-307). ([Link](#))
- **R. Sidibomma** and R. K. Sanodiya, “**Learning Semantic Representations and Discriminative Features in Unsupervised Domain Adaptation,**” 2023 11th International Symposium on Electronic Systems Devices and Computing (ESDC), Sri City, India, 2023, pp. 1-6, doi: 10.1109/ESDC56251.2023.10149872. ([Link](#))